Hypercor 5 mg tablet

Product description
Yellow heart shaped film coated tablet embossed with "S", "5" and breakline on one side, and breakline on the other side. Each film coated tablet contains 5mg of bisoprolol fumarate.

Pharmacodynamics
Bisoprolol is a β1-selective adrenergic blocking agent. Bisoprolol is related structurally to acebutolol, atenolol, and metoprolol in that the drugs contain substituents in the para position of the benzene ring; the presence of large substituents in the para position is believed to account in part for the selective β1-adrenergic blocking effect of these drugs. It is devoid of intrinsic symphathomimetic, membrane-stabilizing properties and only shows shows low affinity to the β2- receptor of the smooth muscle of bronchi and vessels as well as the β2-receptors concerned with metabolic regulation. At therapeutic doses bisoprolol is not to be expected to influence airway resistance and β2-mediated metabolic effects.

Pharmacokinetics
Bisoprolol is almost completely absorbed after oral administration, the absorption is not affected by the presence of food and it undergoes only minimal first-pass metabolism. Peak plasma concentrations are reached 2 to 4 h after oral administration. It is moderately lipid soluble and rapidly and widely distributed in the body. Plasma protein binding is around 30% to 36%. The bioavalability is 82% to 94%. The distribution volume is 3.5 l/kg. It has a plasma elimination half-life of 10 to 12 hours increased in patients with renal or liver impairment. Total clearance is approximately 15 l/h.
Approximately 50% of a dose is metabolised in the liver to pharmacologically inactive polar metabolites that are then excreted by the kidneys; the remaining 50% is excreted unchanged. Bisoprolol is eliminated equally by both liver and renal and less than 2% by feces.

Indication
Hypertension.
Coronary Heart Disease (angina pectoris).
Treatment of stable chronic moderate to severe heart failure with reduced systolic ventricular function (ejection fraction ≤ 35 %, based on echocardiography) in addition to ACE inhibitors, and diuretics, and optionally cardiac glycosides.

Recommended Dosage
Bisoprolol fumarate is given in the management of hypertension and angina pectoris. It is also used as an adjunct to standard therapy in patients with stable chronic heart failure. Treatment should start gradually with low doses and then increased slowly. The dosage should be determined on an individual basis in each case, particularly depending on the pulse rate and the success of treatment.

<u>Hypertension</u>
Bisoprolol is used alone or in combination with other classes of antihypertensive agent in the management of hypertension. The recommended dose is 5 mg bisoprolol fumarate once daily (equivalent to 1 tablet of Hypercor 5mg tablet). As required, the dose may be increased to 10 mg bisoprolol fumarate once daily (equivalent to 2 tablets of of Hypercor 5mg tablet). Further dose increase is justified only in exceptional cases.

<u>Coronary Heart Disease (angina pectoris)</u>
Unless otherwise prescribed, one film-coated tablet of Hypercor 5mg tablet once daily. The maximum recommended dose is 10mg once daily.

<u>Heart failure</u>
Used as an adjunct to standard therapy in patients with stable chronic heart failure. The patients should have stable chronic heart failure without acute failure during the past six weeks and a mainly unchanged basic therapy during the past two weeks. They should be treated at optimal dose with an ACE inhibitor (or other vasodilator in case of intolerance to ACE inhibitors) and a diuretic and optionally cardiac glycosides, prior to the administration of bisoprolol. It is recommended that the treating physician should be experienced in the management of chronic heart failure.

The treatment is to be started with a gradual uptitration from 1.25mg once daily (equivalent to ¼ Hypercor 5mg tablet) and adjusted according to patient's blood pressure response and tolerance. Generally at intervals of 1 – 4 weeks. The maximum maintenance therapy dose is 10 mg once daily (equivalent to 2 Hypercor 5mg tablets).

After initiation of treatment with 1.25 mg, the patients should be observed over a period of approximately 4 hours (especially as regards blood pressure, heart rate, conduction disturbances, signs of worsening of heart failure). The maximum recommended dose is 10 mg once daily. Occurrence of adverse events may prevent all patients being treated with the maximum recommended dose. If necessary, the dose reached can also be decreased step by step. The treatment may be interrupted if necessary and reintroduced as appropriate. During the titration phase, in case of worsening of the heart failure or intolerance, it is recommended first to reduce the dose of bisoprolol or to stop immediately if necessary (in case of severe hypotension, worsening of heart failure with acute pulmonary oedema, cardiogenic shock, symptomatic bradycardia or AV block). Treatment of stable chronic heart failure with bisoprolol is generally a long-term treatment. The treatment with bisoprolol is not recommended to be stopped abruptly since this might lead to a transitory worsening of heart failure. When discontinuing therapy, dosages should be gradually reduced to avoid rebound hypertension. There is no information regarding pharmacokinetics of bisoprolol in patients with chronic heart failure and with impaired liver or renal function. Uptitration of the dose in these populations should therefore be made with additional caution and generally should not exceed 10mg once daily.

<u>Pediatric</u>
There is no paediatric experience with bisoprolol, therefore its use is not recommended for children. Safety and efficacy in children have not been established.

Contraindication
Bisoprolol is contra-indicated in chronic heart failure patients with:
- acute heart failure or during episodes of heart failure decompensation requiring i.v. inotropic therapy
- cardiogenic shock
- AV block of second or third degree (without a pacemaker)
- severe sinus bradycardia
- sinoatrial block
- bradycardia with less than 60 beats/min before the start of therapy
- hypotension (systolic blood pressure less than 100 mm Hg)
- severe bronchial asthma or severe chronic obstructive pulmonary disease
- late stages of peripheral arterial occlusive disease and Raynaud's syndrome
- untreated phaeochromocytoma
- metabolic acidosis
- hypersensitivity to bisoprolol or to any of the excipients
- overt cardiac failure

Mode of Administration
Hypercor 5mg Tablet is administered orally. GI absorption of the drug does not appear to be affected by food. The film-coated tablets are to be taken unchewed together with sufficient liquid, preferably on an empty stomach in the morning or together with breakfast.

Warning and Precautions
Bisoprolol must be used with caution in:
- Anesthesia/surgery (myocardial depression)
- AV block of first degree
- Avoid abrupt withdrawal
- Bronchospasm (bronchial asthma, obstructive airways diseases)
- bronchospastic disease
- Concomitant treatment with inhalation anaesthetics
- Congestive heart failure
- Diabetes mellitus with large fluctuations in blood glucose values; symptoms of hypoglycaemia can be masked
- General anaesthesia
- Hyperthyroidism/thyrotoxicosis
- Liver disease
- Ongoing desensitisation therapy
- Peripheral arterial occlusive disease (intensification of complaints might happen especially during the start of therapy)
- Peripheral vascular disease
- Prinzmetal's angina
- Renal impairment
- Strict fasting

There is no therapeutic experience of bisoprolol treatment of heart failure in patients with the following diseases and conditions:
- NYHA class II heart failure
- Insulin dependent diabetes mellitus (type I)
- Impaired renal function (serum creatinine >300 micromol/l)
- Impaired liver function
- Patients older than 80 years
- Restrictive cardiomyopathy
- Congenital heart disease
- Haemodynamically significant organic valvular disease
- Myocardial infarction within 3 months

The initiation and cessation of treatment with bisoprolol should be a done gradually accompanied by regular monitoring. In bronchial asthma or other chronic obstructive lung diseases, which may cause symptoms, bronchodilating therapy should be given concomitantly). Occasionally, an increase of the airway resistance may occur in patients with asthma, therefore the dose of β2 –stimulants may have to be increased. As with other beta-blockers, bisoprolol may increase both the sensitivity towards allergens and the severity of anaphylactic reactions. Adrenaline treatment does not always give the expected therapeutic effect. Patients with psoriasis or with a history of psoriasis should only be given βbeta-blockers after carefully balancing the benefits against the risks. In patients with phaeochromocytoma, bisoprolol must not be administered until after α-receptor blockade. The symptoms of a thyrotoxicosis may be masked under treatment with bisoprolol.
<u>Effects on ability to drive and use machinery</u>
In a study with coronary heart disease patients bisoprolol did not impair driving performance. However, due to individual variations in reactions to the drug, the ability to drive a vehicle or

to operate machinery may be impaired. This should not be considered particularly at start of treatment and upon change of medication as well as in conjunction with alcohol.

Interaction with Other Medicaments

Combinations not recommended

- Calcium antagonists of the verapamil type and to a lesser extent of the diltiazem type: Negative influence on contractility and atrio-ventricular conduction. Intravenous administration of verapamil in patients on β-blocker treatment may lead to profound hypotension and atrioventricular block.
- Class I antiarrhythmic drugs (e.g. quinidine, disopyramide; lidocaine, phenytoin, flecainide, propafenone): Effect on atrio-ventricular conduction time may be potentiated and negative inotropic effect increased.
- Centrally acting antihypertensive drugs such as clonidine and others (e.g. methyldopa, moxonodine, rilmenidine): Concomitant use of centrally acting antihypertensive drugs may worsen heart failure by a decrease in the central sympathetic tonus (reduction of heart rate and cardiac output, vasodilation). Abrupt withdrawal, particularly if prior to beta-blocker discontinuation, may increase risk of "rebound hypertension".

Combinations to be used with caution

- Calcium antagonists of the dihydropyridine type such as felodipine and amlodipine: Concomitant use may increase the risk of hypotension, and an increase in the risk of a further deterioration of the ventricular pump function in patients with heart failure cannot be excluded.
- Class-lll antiarrhythmic drug (e.g. amiodaron): Effect on atrio-ventricular conduction time may be potentiated.
- Topical beta-blockers (e.g. eye drops for glaucoma treatment) may add to the systemic effects of bisoprolol and have additive effect.
- Parasympathomimetic drugs: Concomitant use may increase atrioventricular conduction time and the risk of bradycardia.
- Insulin and oral antidiabetic drugs: Intensification of blood sugar lowering effect. Blockade of β-adrenoreceptors may mask symptoms of hypoglycaemia.
- Anaesthetic agents: Attenuation of the reflex tachycardia and increase of the risk of hypotension
- Digitalis glycosides: Reduction of heart rate, increase of atrio-ventricular conduction time
- Postagladin synthetase inhibiting drugs: Decreased hypotensive effect.
- Non-steroidal anti-inflammatory drugs (NSAIDs): NSAIDs may reduce the hypotensive effect of bisoprolol.
- β-Sympathomimetic agents (e.g. isoprenaline, dobutamine): Combination with bisoprolol may reduce the effect of both agents.
- Sympathomimetics that activate both β- and α-adrenoceptors (e.g. noradrenaline, adrenaline): Combination with bisoprolol may unmask the α-adrenoceptor-mediated vasoconstrictor effects of these agents leading to blood pressure increase and exacerbated intermittent claudication. Such interactions are considered to be more likely with nonselective β-blockers.
- Concomitant use with antihypertensive agents as well as with other drugs with blood pressure lowering potential (e.g. tricyclic antidepressants, barbiturates, phenothiazines) may increase the risk of hypotension.
- Rifampicin: Slight reduction of the half-life of bisoprolol possible due to the induction of hepatic drug metabolising enzymes. Normally no dosage adjustment is necessary.

Combinations to be considered

- Mefloquine: increased risk of bradycardia
- Monoamine oxidase inhibitors (except MAO-B inhibitors): Enhanced hypotensive effect of the β-blockers but also risk for hypertensive crisis.

Pregnancy and Lactation

Pregnancy:

Bisoprolol has pharmacological effects that may cause harmful effects on pregnancy and/or the fetus/newborn. In general, β-adrenoceptor blockers reduce placental perfusion, which has been associated with growth retardation, intrauterine death, abortion or early labour. Adverse effects (e.g. hypoglycaemia and bradycardia) may occur in the fetus and newborn infant. If treatment with β-adrenoceptor blockers is necessary, β1-selective adrenoceptor blockers are preferable. Bisoprolol should not be used during pregnancy unless clearly necessary. If treatment with bisoprolol is considered necessary, the uteroplacental blood flow and the fetal growth should be monitored. In case of harmful effects on pregnancy or the fetus alternative treatment should be considered. The newborn infant must be closely monitored. Symptoms of hypoglycaemia and bradycardia are generally to be expected within the first 3 days.

Lactation:

It is not known whether this drug is excreted in human milk. Therefore, breastfeeding is not recommended during administration of bisoprolol.

Side Effects

Common:-

Circulation: Feeling of cold or numbness in the extremities.
CNS: Fatigue*, exhaustion*, vertigo*, headaches*.
Gastrointestinal: Nausea, vomiting, diarrhea, constipation.
Metabolism: Increase in triglycerides and cholesterol, hyperglycaemia and glycosuria, hyperuricaemia, disturbances of fluid and electrolyte balance (especially hypokalaemia and hyponatraemia, hypomagnesaemia, hypochloraemia, hypercalcaemia), metabolic acidosis.

Uncommon:-

General: Muscle weakness and cramps.
Circulation: Bradycardia, disturbances of atrioventricular conduction, aggravation of heart failure, orthostatic hypotension.
CNS: Sleep disorders, depression
Airways: Bronchospasm in patients with asthma or a history of airways obstruction.
Renal: Reversible increase in serum creatinine and urea.
Gastrointestinal: Loss of appetite, abdominal pain, increase in amylases.

Rare:-

CNS: Nightmares*, hallucinations.
Skin: Hypersensitivity reactions (pruritis, flushing, rash, photosensitization, purpura, urticaria, erythema).
Liver: Elevation in liver enzymes (ALAT, ASAT), hepatitis, jaundice.
Urogenital: Impotence.
Ear-nose-throat: Impaired hearing, allergic rhinitis.
Eyes: Eye dryness (to be taken into account if the patient wears contact lenses), visual disturbances.
Blood: Leukopaenia, thrombocytopaenia.

Isolated cases:-

Eyes:Conjunctivitis.
Circulation: Chest pain.
Skin: Beta-blockers may induce or aggravate psoriasis or induce a psoriasiform rash, alopecia, cutaneous lupus erythematosus.
Blood: Agranulocytosis.
Gastrointestinal: Pancreatitis.
CNS: central nervous system
ORL: otorhinolaryngology

* These symptoms occur above all at the start of treatment. They are generally mild in severity and usually disappear within 1 to 2 weeks after the initiation of treatment.

Symptoms & Treatment of Overdose

In general the most common signs expected with overdosage of a β-blocker are bradycardia, hypotension, bronchospasm, acute cardiac insufficiency and hypoglycaemia.

If overdose occurs, bisoprolol treatment should be stopped and supportive and symptomatic treatment should be provided. Limited data suggest that bisoprolol is hardly dialyzable.

Based on the expected pharmacologic actions and recommendations for other beta-blockers, the following general measures should be considered when clinically warranted:

- Bradycardia: Administer intravenous atropine. If the response is inadequate, isoprenaline or another agent with positive chronotropic properties may be given cautiously. Under some circumstances, transvenous pacemaker insertion may be necessary.
- Hypotension: Intravenous fluids and vasopressors should be administered. Intravenous glucagon may be useful.
- AV block (second or third degree): Patients should be carefully monitored and treated with isoprenaline infusion or transvenous cardiac pacemaker insertion.
- Acute worsening of heart failure: Administer i.v. diuretics, inotropic agents, vasodilating agents.
- Bronchospasm: Administer bronchodilator therapy such as isoprenaline, β2-sympathomimetic drugs and/or aminophylline.
- Hypoglycaemia: Administer i.v. glucose.

Pack Size

Aluminum-PVC blister strip of 10 tablets per strip. Box of 10 and 50 strips.

Storage Condition

Store in a cool dry place, below 30°C. Protect from light.
Keep out of reach of children. *Jauhi daripada kanak-kanak*.

Shelf life

Please refer to outer package

Manufacturer

SRIPRASIT PHARMA CO., LTD.
216 Moo 6, Suanluang, Krathum Baen, Samut Sakhon 74110, Thailand.

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